

Checksums

This is a block code method where a checksum is created based on the data values in the data blocks to be transmitted using some algorithm and appended to the data. When the receiver gets this data, a new checksum is calculated and compared with the existing checksum. A non-match indicates an error.

Error Detection by Checksums

For error detection by checksums, data is divided into fixed sized frames or segments.

- **Sender's End** – The sender adds the segments using 1's complement arithmetic to get the sum. It then complements the sum to get the checksum and sends it along with the data frames.
- **Receiver's End** – The receiver adds the incoming segments along with the checksum using 1's complement arithmetic to get the sum and then complements it.

If the result is zero, the received frames are accepted; otherwise they are discarded.

Example

Suppose that the sender wants to send 4 frames each of 8 bits, where the frames are 11001100, 10101010, 11110000 and 11000011.

The sender adds the bits using 1s complement arithmetic. While adding two numbers using 1s complement arithmetic, if there is a carry over, it is added to the sum.

After adding all the 4 frames, the sender complements the sum to get the checksum, 11010011, and sends it along with the data frames.

The receiver performs 1s complement arithmetic sum of all the frames including the checksum. The result is complemented and found to be 0. Hence, the receiver assumes that no error has occurred.

Sender's End	Receiver's End
Frame 1: 11001100 Frame 2: + 10101010 Partial Sum: 1 01110110 + 1 01110111 Frame 3: + 11110000 Partial Sum: 1 01100111 + 1 01101000 Frame 4: + 11000011 Partial Sum: 1 00101011 + 1 Sum: 00101100 Checksum: 11010011	Frame 1: 11001100 Frame 2: + 10101010 Partial Sum: 1 01110110 + 1 01110111 Frame 3: + 11110000 Partial Sum: 1 01100111 + 1 01101000 Frame 4: + 11000011 Partial Sum: 1 00101011 + 1 Sum: 00101100 Checksum: 11010011 Sum: 11111111 Complement: 00000000 Hence accept frames.

Advantage :

The checksum detects all the errors involving an odd number of bits as well as the error involving an even number of bits.

Disadvantage :

The main problem is that the error goes undetected if one or more bits of a subunit is damaged and the corresponding bit or bits of a subunit are damaged and the corresponding bit or bits of opposite value in second subunit are also damaged. This is because the sum of those columns remains unchanged.

Explain the Cyclic Redundancy Checks (CRCs)

The Cyclic Redundancy Checks (CRC) is the most powerful method for Error-Detection and Correction. It is given as a **kbit** message and the transmitter creates an $(n - k)$ bit sequence called frame check sequence. The out coming frame, including n bits, is precisely divisible by some fixed number. Modulo 2 Arithmetic is used in this binary addition with no carries, just like the XOR operation.

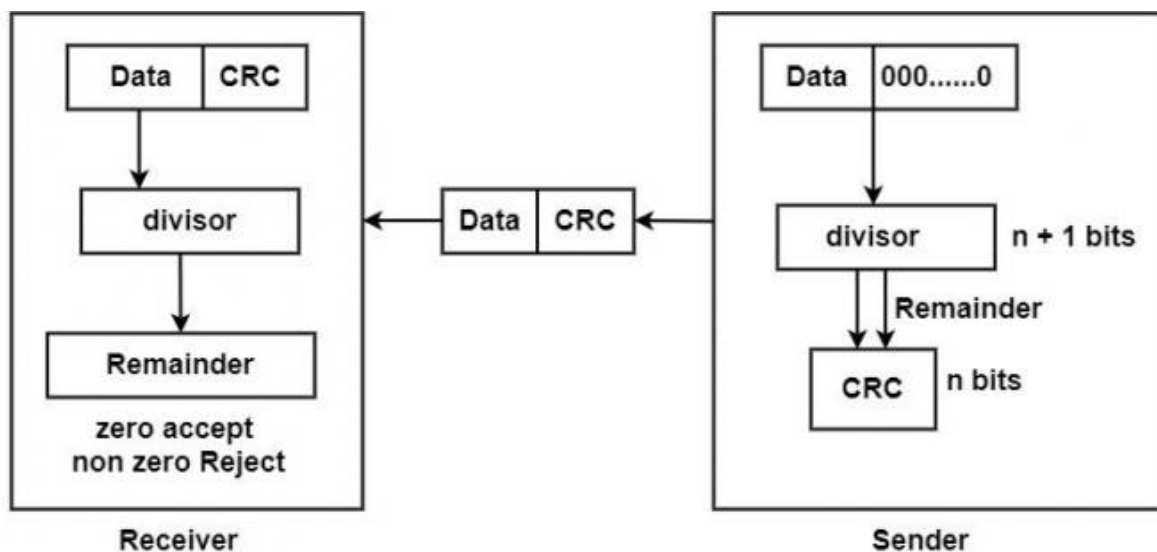
Redundancy means **duplicacy**. The redundancy bits used by CRC are changed by splitting the data unit by a fixed divisor. The remainder is CRC.

- Can detect all odd errors, single bit, burst error of length of the polynomial degree.

Qualities of CRC

- It should have accurately one less bit than the divisor.
- Joining it to the end of the data unit should create the resulting bit sequence precisely divisible by the divisor.

CRC generator and checker



Process

- A string of n 0s is added to the data unit. The number n is one smaller than the number of bits in the fixed divisor.
- The new data unit is divided by a divisor utilizing a procedure known as binary division; the remainder appearing from the division is CRC.
- The CRC of n bits interpreted in phase 2 restores the added 0s at the end of the data unit.

- Divisor can be given as binary or polynomial format.

Example

Message D = 1010001101 (10 bits)

Predetermined P = 110101 (6 bits)

FCS R = to be calculated 5 bits

Hence, $n = 15$ $K = 10$ and $(n - k) = 5$

The message is generated through 2^5 : accommodating 1010001101000

The product is divided by P.

$$\begin{array}{r}
 \begin{array}{r} 1101010110 \end{array} \leftarrow Q \\
 \begin{array}{r} 110101 \overline{) 101000110100000} \\
 \underline{110101} \\
 111011 \\
 \underline{110101} \\
 111010 \\
 \underline{110101} \\
 111110 \\
 \underline{110101} \\
 101100 \\
 \underline{110101} \\
 110010 \\
 \underline{110101} \\
 01110 \leftarrow R
 \end{array}$$

The remainder is inserted to 2^5D to provide $T = 101000110101110$ that is sent.

Suppose that there are no errors, and the receiver gets T perfect. The received frame is divided by P.

$$\begin{array}{r}
 1101010110 \\
 110101 \overline{) 101000110101110} \\
 \underline{110101} \\
 1110111 \\
 \underline{1101101} \\
 111010 \\
 \underline{110101} \\
 111110 \\
 \underline{110101} \\
 101100 \\
 \underline{110101} \\
 110101 \\
 \underline{110101} \\
 0 \leftarrow R
 \end{array}$$

Because of no remainder, there are no errors.